The Illusion of Learning: Toward Growth-Centered AI

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Abstract

This paper argues that educational systems in the age of AI must become behaviorally aware: able to see not just what learners say, but what they do. Struggle is the forcing function of learning, and human tutors calibrate their interventions on observable student effort. However, Large Language Models are fundamentally blind to the behaviors that signal struggle, providing the same response to a student who has tried everything and one who has tried nothing. Prior work shows learners who engage with educational content primarily through AI interaction not only learn less but falsely self-perceive their understanding as higher than it is. In turn, we are creating a generation of learners who do not know they are not learning. To combat this, we must create systems that reward students for what they did, not what they said. To demonstrate the feasibility of behavioral awareness, we deployed an AI-assisted programming environment across eight introductory courses, capturing 882,467 behavioral events from 664 students. We propose a four-layer framework for behavioral awareness and outline a roadmap toward growth-centered AI that responds not just to what learners explicitly say, but to what they implicitly need.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **E-learning**; • **Computing methodologies** → *Natural language processing*; • **Social and professional topics** → *Computing education.*

Keywords

behavioral awareness, productive struggle, AI in education, learning analytics, intelligent tutoring systems, growth-centered AI

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1 Introduction

Struggle is the forcing function of learning. Without struggle, knowledge gain is temporary and fails to transfer [3, 6]. Human tutors know this intuitively, reorienting when and how they intervene based on their students' observable effort, persistence, and frustration [4, 7]. Bloom envisioned a tutor for every learner four decades ago [4], and while Large Language Models are the first technology capable of delivering it, they have arrived pointed in the opposite direction.

In the evolution of educational methods from human tutors to Intelligent Tutoring Systems (ITS) [16] to Large Language Models (LLMs), no technology has offered such frictionless access to information, providing students what they want when they want it. While this low friction is a strength for information retrieval, it is the Achilles heel for learning. Thus, while prior work classifies struggle as either productive or unproductive [9, 10], we argue LLMs require a new type of struggle classification: the avoidance of struggle altogether.

While traditional human tutors reward effort, Large Language Models reward requests. Due to the zero-friction nature of making such requests, learners find themselves in environments that not only fail to reward effort, but are fundamentally blind to it—providing the same response to a student who has tried everything and one who has tried nothing. Even more troubling, LLMs create a facade of understanding; prior work shows AI-dependent students report high understanding while failing to transfer knowledge to similar tasks [2]. With the advent of LLMs, we are creating a generation of learners who do not know they are not learning.

To address these challenges we propose a new class of intelligent assistance centered around the growth of its users. Prior research on LLMs largely focuses on productivity gains, while neglecting the human cost of using such systems. We argue that LLMs should be placed on a spectrum from productivity-focused to growth-focused depending on the context. Movement toward one end comes at the cost of the other. For instance, enacting guardrails that prohibit solution seeking comes at the cost of productivity, while writing all of the code for the user comes at the cost of growth. In educational settings, Large Language Models should orient toward growth. The distinction matters, since Large Language Models are fundamentally highly productive systems; thus, developers must take proactive steps to center the system around the growth of its users [15].

To build growth-centered AI, the system must be behaviorally aware. Since struggle facilitates growth, systems must know the behaviors the user has exhibited and how those behaviors have evolved over time. We show that behavioral context throughout a session provides insights not possible in existing environments—where LLMs typically only see a snapshot of the learner's current context. Behaviorally aware systems enable adaptations to the learner's environment that elicit the conditions for productive struggle, producing growth as a byproduct rather than as an explicit goal.

This paper argues for growth-centered AI and outlines a roadmap to achieve it. Behavioral awareness serves this goal through two primary applications: behavioral intervention, which adapts the

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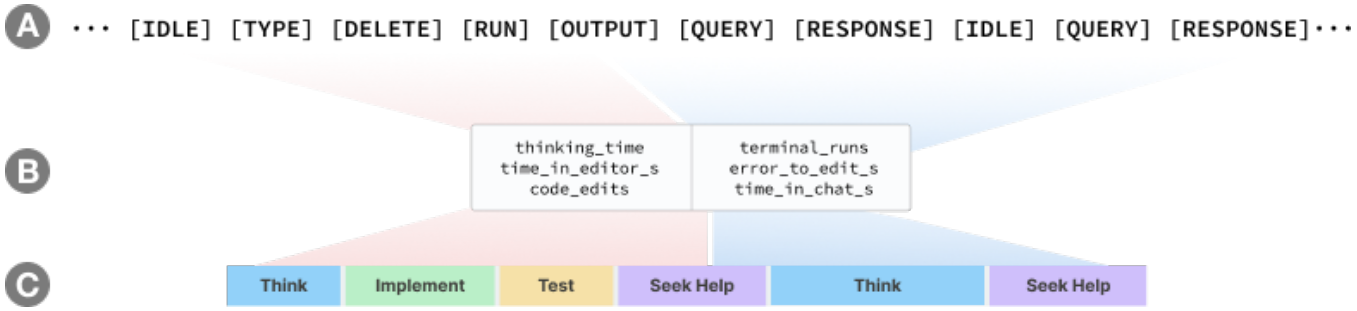


Figure 1: The behavioral awareness pipeline. (A) Raw telemetry captured from the IDE, from which two representations are derived in parallel: (B) observable metrics quantifying effort through event counts, edit frequencies, and activity ratios, and (C) behavioral classification mapping raw event patterns into discrete states such as implementing, debugging, testing, and seeking help.

learner's environment in ways that facilitate productive struggle; and process-focused assessment, which evaluates students by what they did rather than solely by their final output. This new class of AI will lead the way to a beneficial symbiotic relationship between AI and learners, one in which AI responds not just to what users explicitly say but to what they implicitly need.

2 The Cost of Frictionless Assistance

Prior research has studied the role of struggle in learning for decades, consistently showing that deep learning emerges from a behavioral state called productive struggle, often called the Goldilocks zone for learning [9, 18]. Too much struggle can become unproductive and result in disengagement, and the avoidance of struggle comes at the cost of deep learning.

Large Language Models (LLMs) have provided the most substantial disruption to this productive/unproductive struggle model in history. LLMs not only make the avoidance of struggle effortless, but nearly impossible to resist—especially for novice learners—due to the high accessibility and frictionless nature of the interactions. This lack of friction results in the circumvention of the very help-seeking processes that produce learning, such as foraging across resources, formulating a question, and deciding when to ask [14].

2.1 The Facade of Learning

Prior research shows that AI assistance can raise task performance and lower perceived effort while leaving conceptual learning unchanged—a dissociation between performing and learning [1]. Worse, when AI access is subsequently removed, heavy users underperform, indicating reduced skill acquisition [2]. In addition, prior research shows that the effect of AI usage on novice learners is moderated by prior programming experience, suggesting different populations require different levels of intervention [11].

Metacognitive awareness of one's own understanding and learning progress is an extremely difficult skill to develop and largely absent in novice learners [8]. Thus, educational systems must not be reliant on the learner to be the judge of their own understanding; rather, the system should adapt to the understanding the user has exhibited through their actions.

2.2 Guardrails Are Not Enough

The dominant approach to preventing effortless avoidance in interactions with LLMs is through the incorporation of guardrails in the prompt. While prior research has shown that guardrailed tutors can preserve learning where unrestricted LLMs harm it [2], and that such guardrailed tools are viable in real classrooms [13], they are still fundamentally limited by the fact that learner intent going into LLM interactions overrides LLM guardrails. For instance, a learner who wants the solution when going into an interaction will respond to Socratic questioning with increasingly direct and low-effort queries.

More critically, while guardrails constrain the LLM's response, they still fundamentally miss the full context of the learner's input. Without this context, it is impossible for the LLM to appropriately orient itself to meet the learner's needs, as if a human tutor only opened their eyes after the learner asked their question.

3 Behavioral Awareness

Growth is a byproduct of remaining in the environmental conditions that make its arrival inevitable [17]. We argue the behaviors these conditions elicit are detectable and actionable in controlled environments. By detecting and understanding these behaviors, we can shape learner environments toward the conditions that facilitate deep understanding. We formalize this capability as a four-layer *behavioral awareness framework* specifying what systems must capture, classify, abstract, and act upon to facilitate productive struggle.

To achieve growth-centered AI we propose four key layers: (1) raw telemetry as the base, (2) behavioral classification from raw telemetry, (3) abstracting how these behaviors and their intensities evolve over time, and (4) expert-informed interventions that shape learner environments and reorient AI responses based on the learner's exhibited behaviors.

(1) Raw Data. The raw data represents the base layer from which behavioral states and their abstractions are derived. For this layer, developers should capture the full breadth of raw telemetry from their systems, including mouse movements, clicks, keystrokes, LLM interactions, and the source location of each event.

(2) Behavioral Classification. To understand the learner's behaviors throughout their sessions, we classify the raw data into

Pre-Query Effort Groups (N=1,576 queries)

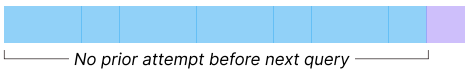
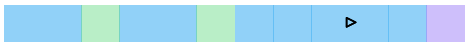
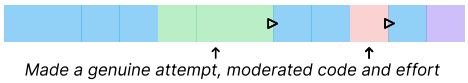
Effort-Level Group	Sample Query	N(%)	Edits	Runs	Dur.	% Dep.
No Effort 	"i do not know bruh, can you tell me"	550 (34.9%)	0.0	0.0	23s	75.6%
Low Effort  *Light effort, often sporadic, with few runs, errors, edits*	"why does it still not work"	261 (16.6%)	5.5	1.6	56s	69.8%
Medium Effort  *Made a genuine attempt, moderated code and effort*	"what am i doing wrong for printing all scores"	508 (32.2%)	38.8	2.4	100s	56.1%
High Effort  *Extensive effort but no resolution with high runs, errors, edits*	"invalid syntax error with line 12"	257 (16.3%)	179.9	7.8	257s	57.0%



Figure 2: Pre-query effort groups (N=1,576 queries from 386 AI using learners). No-effort queries followed zero activity; high-effort queries followed 180 edits and 8 terminal runs over 4 minutes. % Dep. denotes the share of each group's queries classified as dependent help-seeking—offloading cognitive effort to the AI—rather than guided.

behavioral states. Each behavioral state represents a discrete behavior a learner may exhibit while interacting with the system. For instance, in an IDE environment, these might include implementing, debugging, testing, seeking help, and thinking. To identify which behaviors developers should classify in their systems, we recommend observing users interact with the system and noting any behaviors that signal effort or avoidance.

(3) Behavioral Abstraction. To understand how the learner's behavioral states and their intensities evolve over time, we abstract behavioral states into observable metrics and behavioral sequences.

Observable Metrics. Objective measures of learner activity calculated from frequency, ratios, and sums of events. These metrics capture effort levels. For instance, in an IDE environment, insightful metrics might include frequency of code edits, terminal runs, AI interactions, sum total of these events, and delete-to-type ratio.

Behavioral Sequences. The sequential order of behaviors as they occur, providing insights into how long they lasted and how the learner's behavior evolved over the course of the session. This provides the full story of how the learner arrived at each moment of interaction.

(4) Expert-Informed Intervention. Behavioral awareness enables new methods of intervention in educational environments. For instance, adapting the learning environment in ways that modify the learner's behaviors such as changing task difficulty, surfacing hints, or providing targeted guidance such as line highlighting.

In addition, by injecting the learner's real-time behavioral context into the prompt, Large Language Models can see and orient toward what the learner needs as their behavioral states evolve.

Due to the inherent heuristic rules in detection, any prescribed expert intervention method is deployed deterministically, enabling the reliable and accurate programmatic propagation of their expertise at scale. This framework builds on a long lineage of behavioral analysis in learning environments—affect detection from interaction data [5, 7] and help-seeking and metacognitive tutoring in intelligent tutoring systems [12, 14]. That lineage assumed a closed-domain tutor whose interventions were built into the domain model. Our contribution is to generalize behavioral awareness into a design requirement for open-ended LLM environments, where the model that answers the learner is also the surface through which behavior-informed intervention is delivered.

Together, these layers provide systems with the behavioral context needed to understand not just what learners explicitly say, but the story of how they got there [5, 12, 14]. Only then can we as developers create the conditions in our systems that foster deep understanding.

4 Demonstrating Behavioral Awareness

To demonstrate the framework's feasibility, we deployed a task-based IDE with AI tutor integration into eight introductory programming classes at a Carnegie-classified R-1 university in the United States. All deployments were approved by our institution's IRB and students were informed that participation was not mandatory, nor would it affect their course grade.

Across all deployments, 882,467 raw telemetry events were captured from 664 students. Our behavioral classification and abstraction pipeline transformed these raw events into 15,991 behavioral segments and 89,015 observable metrics.

For this IDE environment, we classified the following behavioral states: implementing, debugging, testing, seeking help, and thinking. The classifier's heuristic rules were derived from a team of

four expert labelers annotating student sessions using a behavior codebook; the resulting automated classifier was validated against expert labels and student self-reports, reaching $\kappa = 0.83$ (87% agreement) with expert labelers. Thinking is distinguished from off-task idleness by duration and context: short pauses (≥3 s gaps adjacent to active work) are classified into thinking subtypes based on surrounding activity—reading the task, reviewing code, reading an error, or reading an AI response—while inactivity exceeding 15 seconds is segmented as idle and excluded from effort measures.

To understand how student effort associates with AI usage, we classified queries by pre-query activity level: no effort, low effort, medium effort, and high effort. Figure 2 shows behavioral signatures across these groups.

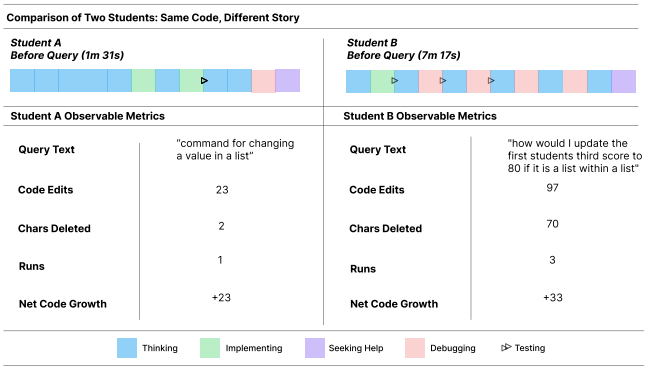


Figure 3: Behavioral sequences preceding two students' first AI query. Both arrived at identical code. Student A wrote 23 edits in 90 seconds. Student B wrote 97 edits, deleted 70 characters, and ran the code 3 times over 7 minutes.

While traditional educational environments provide Large Language Models with the semantic context of the learner's current situation—code content, terminal output, query text—they miss the behavioral story of how the learner got there. For instance, Figure 3 shows two students with drastically different behavioral profiles each arriving at their first query with the same code context:

```
grade_book = [[88,92,75],[70,85,90],[95,88,82]]
grade_book.append([78,84,91])
```

After the session completed, each student was presented with a survey where they reported their self-perceived understanding of the concepts they worked through in the task. For this comparison, each student was assigned to the effort group matching the majority classification of effort across their queries. Figure 4 shows that despite the extreme differences in exhibited effort between no-effort and high-effort students, both reported highly similar levels of perceived understanding.

Prior literature has extensively studied this phenomenon, yet no one has collected the behavioral signals needed to detect these students in real-time. This demonstration does not, however, include a transfer-of-knowledge test, necessary for assessing whether their self-perceived understanding is accurate, which prior work indicates is often not the case for AI-dependent users. In future deployments, as discussed in near-term future work, we plan to conduct knowledge transfer tests for all users. Behavioral signals

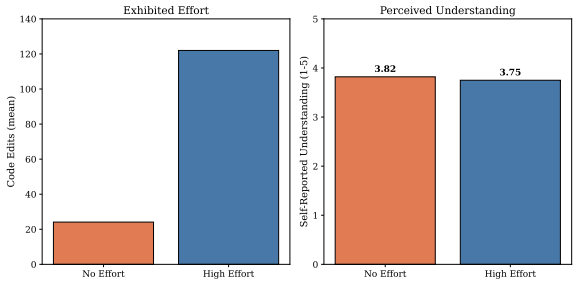


Figure 4: Student-level averages grouped by majority effort classification. Despite large differences in exhibited coding effort, no-effort and high-effort students reported statistically indistinguishable understanding (3.82 vs. 3.75, $p = 0.91$).

are proxies, not ground truth for learners' internal states. Quantity-shaped metrics can misread reflective learners whose effort is invisible to telemetry, and any signal that carries consequences can eventually be performed rather than exercised. Learners can also route around the instrumented environment entirely; our telemetry records when a learner leaves the task tab, but not what happens elsewhere. Process-focused assessment should treat such gaps not as misconduct but as missing evidence—work whose process is unobserved cannot be vouched for by the process record, and its assessment falls back to what the learner can demonstrate without assistance. Thus, behavioral signals should primarily orient systems—serving as evidence for intervention and assessment—rather than as precise classifications of learning.

5 Future Work & Impact

Decades of educational methods have been oriented around guiding learners toward self-discovery, yet with the advent of Large Language Models, institutions have failed to adapt—leaving a generation of learners behind. Thus, as developers, it is our moral obligation to fill this gap by creating systems that help learners resist the forces that obfuscate growth—with the goal of creating generations of learners who can reap the benefits of Large Language Models' unprecedented capabilities while preserving learning itself.

To do so, creating behaviorally aware systems that accurately describe what students did, not just what they said, is a requirement. Widespread adoption of such systems enables a new class of AI systems that can align with human tutors' strength of intervening based on observed behaviors, enabling future educational environments that can more precisely meet the unspoken needs of each student, particularly those who are novice and struggling. We first outline next steps toward a full end-to-end demonstration of our pipeline. Then, we discuss how this new class of growth-centered AI can help shape the future of education in a way that leverages the strengths of Large Language Models while still providing adequate support for learners who use our systems to grow.

5.1 Near-term (0.5–1 years)

We will administer a transfer-of-knowledge test immediately after students complete AI-assisted tasks. While the literature already supports that high AI usage leads to low transfer, we must recreate

this phenomenon to collect the data necessary to understand it. Specifically, we aim to show transfer rates for each effort-level group. We will then test interventions on students who use AI but fail to transfer—those who delegated *custody of verification*, ownership of checking work they did not fully generate, to the assistant—measuring whether targeted support leads to successful knowledge gain. Upon completion, we will open-source all data and pipeline code, enabling future developers to make their systems behaviorally aware and growth-centered.

5.2 Mid-term (2–6 years)

We aim to spread the adoption of behaviorally aware systems to other developers and domains—writing, math, and beyond. To support this, we envision creating an ecosystem where developers and educators can share findings, methods, and intervention strategies. This ecosystem enables the collective intelligence of domain experts to be encoded as deterministic, programmatic interventions—prescribed by communities and propagated to standardized environments in their respective domains. For instance, expert-informed interventions developed for programming education could be deployed across any IDE that adopts the behavioral awareness framework.

5.3 Long-term (10–25 years)

Institutions eventually create their own educational systems with built-in growth-centered AI assistance—not to compete with off-the-shelf productivity-focused AI, but to provide an alternative: systems that enhance learning rather than remove it entirely. To do so, institutions must shift toward process-focused assessment. Product-focused evaluation is no longer viable in the age of AI. We can no longer assess students based on their outputs, but on how they arrived at those outputs—and whether they can be reproduced without assistance. Behavioral awareness makes this shift possible: systems that capture the learner's trajectory provide the evidence needed to assess process, not just product. Deep understanding is valued over surface-level memorization or AI response reading. Understanding is demonstrated through actions. The ultimate vision is to reorient the incentive structure of educational assessment: away from the product alone and toward the process that produced it. The deepest version of this shift designs the signals themselves so that they cannot be performed without being exercised—so that gaming the assessment and doing the learning collapse into the same act. Exams remain the backbone that certifies foundational knowledge; but learning to code is best done by coding, in environments that cultivate deep understanding along the way. If this vision comes to pass, every learner works alongside systems whose eyes are open—watching how understanding is built, not merely what is produced—and the exam survives as the periodic checkpoint certifying what the environment already knows.

6 Conclusion

Large Language Models have presented an unprecedented disruption to the existing education system, yet these systems remain stagnant as a generation of learners is being left behind. As developers, educators, and researchers, we must engage learners in our systems in ways that no longer solely assess output but start assessing the learner's process. To do so, systems must be behaviorally aware. Our deployment across eight programming courses demonstrates that behavioral awareness is feasible. We captured 882,467 events, classified them into behavioral states and effort levels, and revealed a pattern invisible to current systems: students who exerted no effort before querying reported the same understanding as those who struggled extensively. To the LLM, two students with the same semantic context seem identical; only behavioral awareness reveals one struggled and the other did not. This is not a call to abandon AI in education, but rather to adapt AI-integrated systems to the new teaching paradigm. The old way of assessing students by what they produce is dead; the only way forward is by what they did. Behavioral awareness enables this paradigm, opening the possibility for new methods of interaction that bring learners toward the conditions that facilitate their understanding, not obfuscate it.

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